

FSE Assignment

Take-Home: Buildable Land Analysis

What we're after

Before anyone builds on a piece of land, they need to know how much of it is actually usable. A parcel might be 100 acres on paper, but once you take out regulatory setbacks and physical or environmental constraints - wetlands, the 100-year floodplain, transmission-line easements, existing buildings - the real buildable area might be 60 acres, or none.

We'd like a small full-stack app that, given a parcel and a set of constraint layers, works out the buildable area (with a breakdown of what was removed), shows the result on a map, and lets someone adjust it by hand.

In plain terms: take a piece of land, take away the bits you legally or physically can't build on, and tell me how much usable land is left on a map I can poke at.

What to build

- A backend that takes a parcel and a few constraint layers and returns the buildable area plus a short breakdown of what was removed and why.
- A map that shows the parcel with buildable vs. excluded areas, and lets the user draw an area to exclude (carve out) and draw an area to add back (restore) — with the buildable total and the breakdown updating as they go.
- A short writeup of how you approached it and the calls you made.

Suggested stack: a Python / FastAPI backend and a React frontend with a map (for example the ArcGIS Maps SDK for JavaScript, or MapLibre). Use something else if it genuinely helps — just tell us why, especially if the reason is performance.

Data

Use public data — nothing paid. Two layers to get you going:

Layer	Where to get it
Parcels	A Texas county from TNRIS — https://data.tnris.org (pick a county with a manageable parcel count).
Wetlands	USFWS National Wetlands Inventory —

	https://www.fws.gov/program/national-wetlands-inventory/wetlands-data
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Bring in any other constraint layers you think matter and can find — for example FEMA flood zones, transmission lines, building footprints, or protected areas. Deciding which constraints are worth modeling, and sourcing them, is part of the exercise.

Setbacks

Most of these constraints aren't simply “avoid the polygon” — they usually come with a setback or buffer (you keep some distance from a wetland, an easement, a structure, and so on). Choose sensible default distances for whatever you model, and tell us where they come from. We care more about your reasoning and your sources than about an exact figure.

Make those distances **configurable** - we should be able to change, say, the wetland buffer or an easement width and re-run without editing code (a config file, request parameters, or controls on the map your call).

A few basics

- It runs from a clean checkout with a short README.
- The map is interactive (pan, zoom, click), and buildable vs. excluded is clear.
- The totals add up.
- It works on real data, which is messier than a tutorial.

Performance

It should feel responsive and handle a realistic amount of data, not just one tiny parcel. Tell us how it behaves as the data grows and where it would start to strain.

What to send

- A repo we can run.
- A short writeup: your approach, the main tradeoffs, your data and setback choices with sources, and where it breaks or what you'd do next.